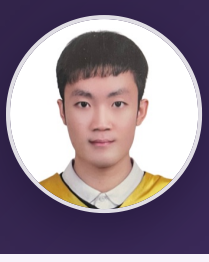
AI Application R&D Engineer · BenQ Materials

Shan-Cheng Chang 張善程

LLM Agents · Knowledge Graph · Document AI

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— CORE SKILLS

LLM & Agents
Prompt Engineering · RAG · Knowledge Graph · LLM Agents · Harness Engineering · MCP · Azure OpenAI · Gemini API

Document AI / CV
Azure Document Intelligence · OCR + LLM Extraction · YOLOv8 · Mask R-CNN · PyTorch · OpenCV

Data / ML
XGBoost · SHAP · Neo4j · Qdrant · pgvector

Backend & Web
Python · FastAPI · Node.js · React · TypeScript · SQL Server · PostgreSQL · Firestore

Cloud & DevOps
Docker · GCP (Firebase, Cloud Run) · Azure AI · Git

Security
AES-256-GCM · JWT · AD / LDAP · Entra ID OAuth · RBAC · ISO 27001

— EDUCATION

National Taiwan University
M.S. Biomechatronics Eng. (AIoT / CV) · 2021 – 2023

National Taipei University of Technology
B.S. Mechanical Engineering · 2017 – 2021

— LANGUAGES

Mandarin Chinese — Native
English — Professional (TOEIC 790)
German — A2
Military service completed

AI application engineer with 2 years of shipping production LLM systems in manufacturing. Lead developer of BenQ Materials' Enterprise Brain — a knowledge graph plus a Claude Code–style agent harness, productized as an on-prem AI data-analysis platform for a production line — after building the company's document-AI stack from zero (six OCR + LLM systems live across finance, procurement, HR and customs). Owns the full loop from user interviews to deployment and adoption.

6 AI SYSTEMS IN PRODUCTION	**97%+** AVG. OCR FIELD ACCURACY
1,500+ h SAVED PER YEAR	**~1M NTD** ANNUAL COST BENEFIT

— EXPERIENCE

AI Application R&D Engineer
Dec 2024 – Present

BenQ Materials Corp. (明基材料) · Digital Technology Center, Intelligent Application Dept., Intelligent Application Development Sec. · Taoyuan, Taiwan

- **Enterprise Brain → AI Data Analysis Platform** (lead developer) — knowledge graph (Neo4j + PostgreSQL) and a Claude Code–style agent harness, productized as the on-prem AI agent for the polarizer line: MCP tools, sandbox, audit log, 6 n8n workflows → skills, XGBoost yield model (F1 0.74).
- **Document-AI applications (OCR + LLM)** — built the company's platform from zero (Azure Document Intelligence + Azure OpenAI, Python/FastAPI, React, SQL Server) and the OCR → LLM → validation → feedback pipeline every later project was cloned from; six systems live across finance, procurement, HR and customs.
  - **Quotation OCR** — accuracy **51% → 100%**, 3× faster, 758+ documents; automated procurement audit pre-check with explainable bounding boxes.
  - **Taiwan & China invoice OCR** — **97% / 100%** accuracy incl. handwritten; saves **1,447 h/yr** and avoids 480–760K NTD/yr outsourcing; live at 3 sites.
  - **Health-check OCR platform** — 48 medical fields at **97.4%**; AES-256-GCM encryption, AD + JWT, RBAC and audit logging to ISO 27001.
  - **APO-OCR prompt-optimization engine** — designed the golden-set + code-validator + coach-LLM loop; mentored an intern to ship it (**8/8** held-out for $1.44).
- **Google Cloud Platform — supplier ESG questionnaire system** — React 19 + Cloud Functions + Firestore; 33K LOC, 52 APIs, delivered solo in 4.5 months; Entra ID login saved ~10K NTD/user.

Shift Supervisor (值班股長)
Jan 2024 – Nov 2024

China Steel Corporation (中國鋼鐵) · Cold-Rolling Mill · Kaohsiung, Taiwan

- **Ran shift operations** on the cold-rolled coil line, resolving anomalies to keep production continuous at **95% yield**.
- **Data-driven anomaly analysis** — consolidated process parameters (rolling force, tension, speed, temperature) with quality-inspection data; used SPC, trend and correlation analysis to find root causes of coil defects and downtime, proposed process adjustments and tracked their effect.
- **Anomaly statistics & defect-classification reports** (Python / Excel) for daily production meetings and cross-team decisions with upstream/downstream lines and QA — the process-data analysis that later became the domain base for the production-line AI analytics platform at BenQ Materials.

— RESEARCH & SELECTED PROJECTS

AIoT Pest Monitoring & Bee-Pollination Optimization (M.S. thesis, NTU)
2021 – 2023

Lead developer of greenhouse AIoT systems: YOLOv8 + StrongSORT bee tracking (*99.5% mAP*), Mask R-CNN pollen segmentation, deep-learning pest counting (*95.8%*), kriging-based climate control.

PyTorch · YOLOv8 · StrongSORT · Mask R-CNN · IoT · Kriging

Universal OCR Platform — no-code OCR workflow builder (side project)
2025 – 2026

Self-serve platform: define fields and prompts, run a 7-step OCR → LLM pipeline, deploy as a REST API; multi-file joint extraction; three cloud deployment variants.

FastAPI · React 19 · PostgreSQL / Supabase · Cloud Run · Firebase Hosting · Docker

Autonomous Navigation with Autoware / ROS (B.S. capstone, NTUT)
2020 – 2021

Indoor/outdoor self-driving platform on Ubuntu + ROS + Autoware (mapping, localization, path planning) with an ML mask-detection module.

— PUBLICATIONS & PATENT

Chou, C.-Y., **Chang, S.-C.**, et al. (2023). Development of AIoT system for facility asparagus cultivation. *Computers and Electronics in Agriculture*, 206:107665. *(co-first author with advisor)*

Chang, S.-C., et al. (2022). AIoT based pest counting system for asparagus cultivation. *Proc. 10th ISMAB.*

Taiwan Invention Patent TW I842416 B — Crop Harvester, 2024.